CRIME PREDICTION AND ANALYSIS: UNCOVERING PATTERNS AND BUILDING A PREDICTIVE MODEL

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled CRIME PREDICTION AND ANALYSIS: UNCOVERING PATTERNS AND BUILDING A PREDICTIVE MODEL submitted by Fazna Fathima Shaju (Reg. No: 213222) in partial fulfilment of the requirements for the award of Master of Science in Computer Science with Specialization in Data Analytics is a bonafide record of the work carried out at Kerala University of Digital Sciences Innovation and Technology under our supervision.

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DECLARATION

I, FAZNA FATHIMA SHAJU, student of Master of Science in Computer Science with specialization in Data Analytics, hereby declare that this report is substantially the result of my own work, except where explicitly indicated in the text, and has been carried out during the period February 2023 to June 2023.

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ABSTRACT

Crime prediction and analysis are essential components of maintaining public safety and enabling proactive measures in resource allocation and decision-making. By using the strength of ML and DL techniques, combined with in-depth data analysis, this project intends to push the frontiers of crime prediction and analysis. Specifically, the study focuses on four major crime categories: theft, battery, criminal damage, and drugs.

To lay the foundation for accurate crime prediction, the project commences with meticulous data preprocessing. This step involves eliminating irrelevant columns that do not contribute significantly to the prediction task and handling missing values appropriately. Ensuring data quality is of utmost importance, and techniques such as scaling and outlier removal are employed to maintain the integrity and reliability of the dataset.

Feature engineering techniques serve as a crucial tool in extracting meaningful information from the available data. In this project, these techniques are utilized to uncover valuable temporal insights. By converting the date column into more granular components such as month, time, and season, a deeper understanding of crime trends and patterns is achieved. Temporal features provide researchers and law enforcement agencies with valuable insights into the dynamics of crime occurrences.

Exploratory data analysis (EDA) serves as a vital step in unraveling the complex relationships within the dataset. Through EDA, researchers delve into crime type counts, examining the distribution of crimes across different time periods and locations. Furthermore, the in-depth analysis explores the connection between crime types and various factors such as arrests and incidents of domestic

violence. By constructing a correlation matrix, interdependencies among different variables are revealed, shedding light on the intricate nature of crime and its underlying factors.

The outcomes of the project highlight the effectiveness of ensemble learning techniques in accurately classifying different types of crimes based on historical data. Ensemble learning leverage the collective power of multiple models to provide robust predictions and improve overall accuracy. By combining the strengths of individual models, the ensemble approach achieves superior performance in crime prediction tasks, enabling law enforcement agencies to allocate resources efficiently and develop targeted crime prevention strategies.

The insights derived from EDA, coupled with the powerful ensemble learning models, offer practical implications for law enforcement agencies. The framework and models developed in this project enable agencies to make informed decisions regarding resource distribution, crime prevention tactics, and decision-making processes. By leveraging the predictive power of ML and DL techniques, law enforcement agencies can proactively address crime hotspots, allocate resources effectively, and ultimately enhance public safety.

In conclusion, this study effort makes a substantial contribution to the arena of crime analysis and prediction by offering a thorough look of crime patterns and trends. The findings and recommendations derived from this study enhance the understanding of crime dynamics and support data-driven decision-making for law enforcement organizations. To further advance the field, future research can explore the integration of additional data sources, investigate cutting-edge feature-engineering strategies, dimensionality reduction techniques, and anomaly detection algorithms. Additionally, the exploration of more sophisticated deep learning techniques holds promise for identifying rare or atypical crime patterns. These advancements will contribute to even more accurate and effective crime

prediction and analysis, thereby fostering safer communities and enhancing societal well-being.

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Glossary

EDA	Exploratory Data Analytics
ML	Machine Learning
DL	Deep Learning
AUC	Area Under the Curve
ROC	Receiver Operating Curve
SVM	Support Vector
KNN	K-Nearest Neighbour
XG Boost	Extreme Gradient Boosting
MLP	Multi Layer Perceptron
CNN	Convolutional Neural Network
LSTM	Long Short Term Memory
RNN	Recurrent Neural Network

Chapter 1

Introduction

Communities across the world confront a serious problem with crime, and efficient law enforcement techniques may greatly benefit from the capacity to anticipate and examine crime trends. The goal of this research is to use ML Algorithms to analyse past crime data and create models that can reliably forecast when crimes will occur [1]. The report gives a general outline of the project's goals and methods.

1.1 Background and Motivation

Crime poses risks to public safety, societal stability, and individual wellbeing and is a continuous problem for all communities. Law enforcement organisations are increasingly relying on data-driven strategies that make use of cutting-edge tools and analytical methods to solve this problem. In particular, prediction of crime and analysis have become important tools for comprehending crime trends, locating high-risk regions, and allocating resources as efficiently as possible [3].

The urgent need to create reliable crime model predictions that may aid law enforcement organisations in their attempts to prevent crime serves as the driving factor behind this research. Authorities may proactively allocate resources, improve monitoring, and carry out targeted interventions by properly predicting the sorts of crimes that will occur and comprehending the underlying causes that contribute to criminal behaviours. These preemptive steps might lower crime rates, boost public safety, and enhance communities' general quality of life.

1.2 Objectives

The main goals of this initiative are:

1.2.1 Exploratory Data Analysis (EDA):

To perform an in-depth evaluation of this crime dataset, looking at temporal patterns, relationships across variables, and significant takeaways. EDA enables a greater comprehension of crime trends, connections between different crime categories, and the effects of variables like arrests as well as domestic violence episodes.

1.2.2 Crime Prediction Model Development:

To generate machine learning models that can correctly forecast different sorts of crimes based on past data. Our goal is to develop models that can derive information from historical crime patterns, recognise pertinent elements, and produce precise forecasts for upcoming crimes. We will do this by utilising the power of cutting-edge algorithms.

1.2.3 Visualization Dashboard:

Creating a user-friendly dashboard that allows interactive study of crime data and visualises arrests, domestic violence incidences, and their relationships with seasonal and nocturnal variables. The dashboard supports data-driven decision-making and offers a thorough picture of crime trends.

1.2.4 Recommendations for Law Enforcement Agencies:

To aid law enforcement agencies in making judgements by offering analysis- and model-based insights and suggestions. Authorities may adopt targeted crime pre-

vention tactics, improve public safety, and optimise resource allocation by identifying high-crime locations, comprehending temporal trends, and spotting linkages.

1.3 Scope and Limitations:

This project's scope includes the study and forecasting of different crime subtypes for a given period of time and location. The dataset utilised contains drug, theft, violence, and criminal property damage offences. It is vital to recognise the fact that crime predicting is a complicated endeavour impacted by multiple economic, social, cultural, and environmental aspects, even though efforts have been made to assure data quality and model performance. As a result, it might not always be possible to forecast crimes with complete accuracy. The efficiency of the generated models may also change when used in other locations or at various times. We will go into detail about the methods used in the next sections, including data pretreatment, feature engineering, exploratory data analysis, model construction, assessment, and the creation of the visualisation dashboard. To provide a thorough grasp of crime prediction and analysis and its possible influence on law enforcement techniques, the project's findings, learnings, and future implications will be given in depth.

Chapter 2

Literature Review

The ability of crime analysis and prediction to support law enforcement organisations in making proactive decisions and allocating resources has attracted a lot of interest in recent years. To create precise and effective crime prediction models, both scholars and practitioners have investigated a variety of strategies and procedures. The important research and methodologies in the topic are outlined in this review of the literature, along with their advantages and disadvantages.

2.1 Traditional Statistical Approaches:

In the beginning, statistical techniques including regression analysis, time series analysis, and spatial analysis were the main emphasis of crime prediction research. With these methods, it was hoped to find relationships between crime rates and socioeconomic variables, regional traits, and temporal patterns. For instance, research has shown that places with high population density and low income levels typically have greater crime rates. Although these techniques gave useful information on crime trends, they sometimes lacked the capacity to identify intricate linkages and make forecasts in real time [6].

Crime analysis and prediction are essential for improving crime prevention tactics. Researchers have investigated techniques including support vector machines, multivariate time series, and artificial neural networks despite difficulties in reliably forecasting future crimes. However, it is still difficult to anticipate where crimes will occur. In order to determine current implementations and suggest improvements for effective crime analysis and pattern detection. To get over these issues and boost the precision of crime prediction techniques, more research is required [11].

[15]Investigates the use of data analysis techniques such association mining, clustering, classification procedures, correlation, and regression in order to identify

the relationship between crime and societal development for the aim of crime prevention. The study uses the Weka and R tools to preprocess and apply these algorithms to data collected by the government. Several learning strategies are utilised to anticipate crime rates in various states in order to improve the prediction accuracy of regression models. The project provides informative data in an effort to assist local police agencies in lowering crime.

2.2 Machine Learning Techniques:

The goal of research [9] is to employ data science and machine learning methods to forecast crime using the Chicago crime dataset. To achieve high accuracy, the data is preprocessed, including feature selection and scaling. Testing is done on algorithms like K-Nearest Neighbour (KNN) classification, and the most precise one is then utilised for training. In order to analyse crime trends depending on time and month, visualisation techniques are used [8]. The project's goal is to show how machine learning may be used to detect and prevent crime in other places with access to the datasets outside of Chicago.

Using machine learning techniques including Naive Bayes, SVM, Linear Regression, Decision Tree, Bagging Regression, Stacking Regression, and Random Forest, the primary goal of [7] is to anticipate crime in India. In terms of prediction power and classification accuracy, the proposed technique outperforms prior models. The results demonstrate how criminological theories and actual crime data may coexist, and the suggested technique is effective in foretelling future crimes.

Machine learning techniques (K-means, Naive Bayes, and Linear Regression) are used in[9] to forecast the state with the greatest crime rate. In order to anticipate and analyse crime rates, the project aims to provide effective machine learning categorization techniques. Based on efficacy and accuracy, the best algorithm is selected to assist the public in finding safe routes and detecting crime-prone areas [7]. In the proposed system, which employs a number of algorithms, decision trees outperform Naive Bayes for classification accuracy, with SVR regression proving to be the most suitable for prediction.

The primary emphasis of this study is on analysing and forecasting crime patterns using data from Kaggle [6].In compared to past attempts, the Nave Bayes technique achieves a high level of accuracy when categorising different crime patterns. Two classifiers, Multi-nominal NB and Gaussian NB, are able to handle the challenges that come with dealing with nominal and real-valued attributes[13]. The naive bayesian classification approach may be used to predict and pinpoint the most common crimes. Evaluation metrics like as precision, recall, F1 score, and accuracy are used to assess an algorithm's efficacy.

2.3 Ensemble Learning:

The ability to mix many models and increase prediction accuracy has made ensemble learning techniques like Random Forest, is prominent in the field of crime prediction. By using the advantages of each model, these strategies develop models in a sequential manner with an emphasis on reducing prediction errors. Using ensemble approaches, it is possible to detect and predict crime hotspots and high-crime locations with reasonable accuracy [2].

2.4 Deep Learning and Neural Networks:

CNN, RNN and other deep learning approaches, in particular, have excelled in a number of fields, including the prediction of crimes [10]. CNNs are useful for extracting characteristics from visual data or images connected to crimes, which helps with tasks like helping to identify criminals in surveillance film. RNNs may be used to examine temporal dependencies and forecast changes in crime trends over time. LSTM (Long Short-Term Memory) networks, for instance, have been used to anticipate crime rates at various time scales [5].

2.5 Spatial and Temporal Analysis:

Theft, violence, drug usage, and criminal damage in Chicago are the main topics of this essay. To forecast the spatial distribution of crime, it uses machine learning algorithms and visualisation technologies. The study examines criminal symbiosis and analyses the geographical and temporal aspects of crime. It also visualises aggregated data and crime dispersion heat maps. The research offers insights into crime patterns and assists in creating efficient crime prevention methods by fusing daily activities theory with crime distribution theory [14].

[12] Discusses integrating a Safe Route Travel app and a Crime Mapping system using a crime prediction model based on the Fuzzy K-Nearest Neighbour algorithm. The model predicts a location's vulnerability to various types of crimes based on historical crime data. The technology aims to improve road safety, shorten wait times between the public and law enforcement, and support the study of crime patterns. The model displays astounding accuracy and may be incorporated into various platforms. Future ideas include collaborating with the cyber crime agency and utilising global data to enhance criminal mitigation techniques.

2.6 Data Integration and Feature Engineering:

By combining sentiment polarity and meteorological variables, [4] seeks to improve crime prediction. It examines Twitter data for sentiment analysis, determines important weather indicators, uses kernel density estimation for crime distribution, and then combines these aspects to create a prediction model. In comparison to typical models, the study's findings show a link between crime, weather, and attitude. Future study can concentrate on shorter time-interval weather data and investigating certain crime-sentiment correlations with cutting-edge techniques.

Through the use of language analysis and statistical topic modelling, [4] research investigates the use of Twitter data for crime prediction. Twitter themes were included, which enhanced crime prediction for different sorts of crime. The study pinpoints performance stumbling blocks and recommends new topics for investigation, including semantic analysis and temporal modelling. The study has consequences for how resources are allocated for preventing crime and for comprehending the geographic locations of Twitter users.

Table 2.1: Summary of Papers

Research Paper	**Methodology**	**Model**	**Accuracy**
[6]	Data preprocessing, Naïve Bayes analysis, feature labeling	Naïve Bayes (Bernoulli NB, Gaussian NB)	93.07%
[3]	Machine Learning Techniques	Naive Bayes, SVM, Linear Regression, Random Forest Regression	99.9%
[2]	N-ensemble learning, statistical data visualization	Naive Bayes, J48, Random Tree, 1-ensemble, 3-ensemble	82.02%, 81.61%, 79.24%
[9]	Crime rate prediction, efficiency and accuracy comparison	K-means, Naive Bayes, Linear Regression, Decision Trees	44.28%, 90%, 100%
[12]	Combining Safe Route Travel app, Fuzzy K-Nearest Neighbor	Fuzzy KNN (FKNN), KNN model	79% to 92.55%
Proposed Method	Data preprocessing, EDA, model selection, Hot Spot Mapping, dashboard creation	SVM, KNN, Random Forest, XG Boost, Logistic Regression, Naive Bayes, Gradient Boost, MLP, SVM with Kernel Trick	81% (XG Boost)

Chapter 3

Materials and Methods

3.1 Data Collection and Preprocessing:

The Chicago Police Department maintains and often updates the Chicago Crime Dataset, which was obtained via Kaggle for this study. The dataset, consisting of approximately 10,000 entries, was deemed relevant and suitable for the research project. It provided valuable information regarding various crimes in the city of Chicago, forming the basis for the analysis and predictions conducted in this study.

Table 3.1: Major Attributes of Crime Dataset

Attribute	**Description**
IUCR	IUCR code
Primary Type	Primary crime type
Description	Description of the crime
Location Description	Description of the crime location
Arrest	Indicator of whether an arrest was made
Domestic	Indicator of whether the crime involved domestic violence
Beat	Police beat where the crime occurred
District	Police district where the crime occurred
Ward	Ward where the crime occurred
Community Area	Community area where the crime occurred
FBI Code	FBI crime code
X Coordinate	X-coordinate of the crime location
Y Coordinate	Y-coordinate of the crime location
Year	Year of the crime occurrence
Latitude	Latitude coordinate of the crime location
Longitude	Longitude coordinate of the crime location
Location	Latitude and longitude coordinates of the crime location

date	iucr	primary_ty...	arrest	domestic	location
06/12/2018 08:41:00	0110	HOMICIDE	True	False	(41.704419193, -87.624304349)
09/20/2012 11:13:00	0110	HOMICIDE	False	False	(41.701624878, -87.629117507)
12/17/2016 01:25:00	0110	HOMICIDE	True	False	(41.703478083, -87.626563168)
09/29/2005 04:20:00	0265	CRIMINAL SEXUAL ASSAULT	True	False	(41.705113794, -87.623461757)
12/21/2007 11:00:00	0266	CRIM SEXUAL ASSAULT	False	False	(41.700950796, -87.61115238)
01/01/2019 09:00:00	0266	CRIMINAL SEXUAL ASSAULT	False	False	

Figure 3.1: Chicago Crime Dataset

Unnecessary columns like "Block," "Description," and "Location Description" were removed as part of the preprocessing process since they had minimal impact on crime prediction. Columns with titles like "Arrest" and "Domestic" were changed from boolean values to binary values to make analysis more straightforward. The dataset was cleaned of duplicates, and the proper handling of missing values was done.

3.2 Feature Engineering:

The given data was processed using feature engineering techniques to extract relevant information. In order to record the month and time components, the "time" and "month" columns were introduced along with a date-time format change to the "date" column. After the seasons were established based on the month, a new column called "season" was introduced. The 'hour' component was divided into morning, noon, and night to form the 'hourcategorized' column. Then, for the purpose of modelling, numerical representations of the "season" and "hourcategorized" columns were provided.

In consideration of low frequency, several entries were chosen to be excluded from the "Primary Type" column during the data preparation stage. Within the 'Primary Type' column, only the following four classes were left: Theft, Criminal Damage, Battery, and Narcotics. The number labels for these classes - Mapping of Class Labels, were generated in a table to help in interpretation and analysis.

Dataset before (Figure 3.1) and after (Figure 3.2) feature engineering and data preprocessing is given in the respective Figures.

Table 3.2: Mapping of Class Labels

Class	Class Label
0	BATTERY
1	CRIMINAL DAMAGE
2	NARCOTICS
3	THEFT

	date	time	month	season	hour	hour_categorized	Primary Type	Arrest	Domestic	Beat	Year
5	2012-11-01	10:00:00	11	0	10	0	THEFT	0	0	2522	2012
26	2012-01-01	00:00:00	1	3	0	2	BATTERY	0	0	122	2012
36	2012-12-01	00:01:00	12	3	0	2	THEFT	0	0	121	2012
48	2012-09-01	00:00:00	9	0	0	2	BATTERY	0	0	1024	2012
86	2012-01-05	08:00:00	1	3	8	0	THEFT	0	0	1915	2012
...	...	...	...	...	...	...	...	...	...	...	...
336237	2012-04-14	20:20:00	4	1	20	2	BATTERY	0	0	312	2012
336238	2012-06-12	09:00:00	6	2	9	0	THEFT	0	0	421	2012
336247	2012-07-21	02:30:00	7	2	2	2	BATTERY	0	0	223	2012
336250	2012-04-30	23:10:00	4	1	23	2	BATTERY	1	1	1112	2012
336254	2012-07-11	03:27:00	7	2	3	2	BATTERY	1	0	624	2012

205945 rows × 11 columns

Figure 3.2: Dataset after preprocessing and Feature-Engineering

3.3 Check for Seasonality:

A time series decomposition approach was used to look for seasonality in the crime data. In order to do this, the time series has to be broken down into its trend, seasonality, and residual components. The Time Series Decomposition was carried out to find any notable repeating patterns or seasonal fluctuations in the incidence of crime throughout the course of the examined period. By assessing the seasonality component, it was determined whether the data showed a consistent and predictable trend throughout numerous seasons. This process was essential to comprehending the seasonal pattern of crime occurrences and determining whether season-specific modelling techniques were required. Because there were no notable seasonality patterns, it was possible to use more generic forecasting methods in the subsequent modelling phase because the crime data did not show major seasonal fluctuations.

3.4 EDA:

To identify the connections between variables and obtain insights into the dataset, EDA was used. To determine the predominate categories, the number of crimes of

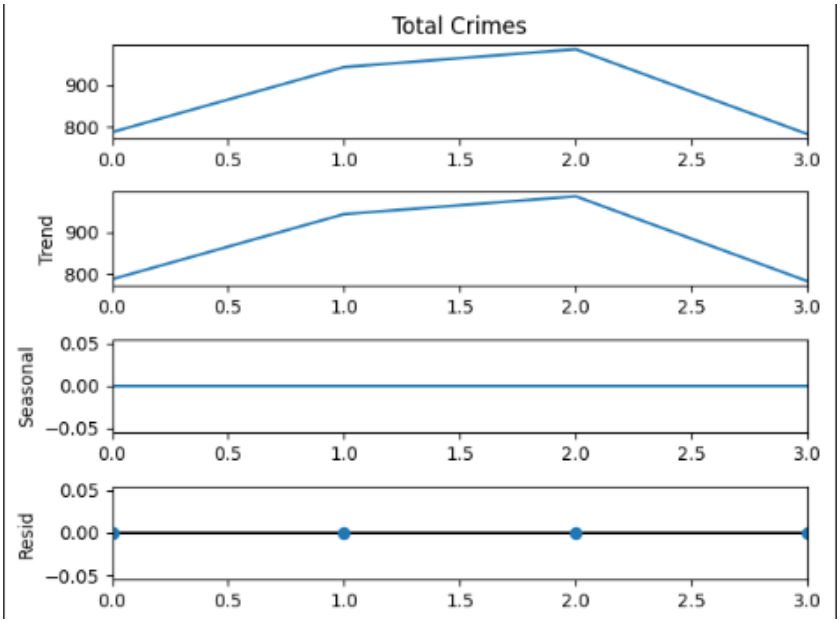

Figure 3.3: Seasonality check using Time Series Decomposition

each category was evaluated. To identify temporal trends, the distribution of criminal incidents by hour and season was examined. To find probable correlations, relationships between crime categories, arrests, and domestic violence were examined. The relationships between the numerical variables were calculated using a correlation matrix.

3.5 Model Selection and Approach:

This section's emphasis was on choosing and using suitable models for crime prediction. The aim was to efficiently categorise crime data into several categories using a combination of classification models, ensemble models, and a deep learning (DL) model. The effectiveness and precision of each model's ability to forecast crime trends were assessed. The models chosen span a wide range of methods, each with advantages over the others in handling certain elements of crime prediction. The best models were found through this procedure to improve the project's crime prediction's efficacy and accuracy. Three types of models were utilized: classification models, ensemble models, and a deep learning (DL) model. Each model was trained and evaluated for its effectiveness in crime prediction.

3.5.1 Classification Models:

Classification models are used to assign data instances to predefined categories or classes. In this project, the following classification models were employed:

SVM:

SVM is a potent method that identifies the ideal hyperplane to divide data points into several groups. Effective categorization is achieved by maximising the gap between the classes.

KNN:

Based on how similar they are to neighbouring data points, data points are classified using the simple yet powerful KNN algorithm. A new data point is added for the class with the highest representation among its k nearest neighbours.

Logistic Regression:

For binary classification, a common approach is logistic regression. It is suitable for forecasting probabilities because it uses the logistic function to represent the connection between the dependent variable and independent factors.

Naive Bayes:

The Naive Bayes algorithm uses the Bayes theorem and is a probabilistic method. By using the feature probabilities and assuming that the characteristics are independent, it determines the likelihood that an instance belongs to a certain class.

SVM with Kernel Trick:

The kernel technique combined with SVM is an effective classification model. It can capture nonlinear relationships in the data by using the kernel method, making it suited for challenging crime prediction applications. By employing hyperplanes to divide data points into classes, SVM increases the margin between classes. The accuracy of crime prediction in this study is improved by its strong performance and effectiveness in processing high-dimensional data.

3.5.2 Ensemble Models:

Ensemble models combine multiple individual models to improve prediction accuracy. In this project, the following ensemble models were employed:

Random Forest:

A number of decision trees are constructed as part of the Random Forest ensemble learning approach, and their predictions are then integrated to provide the final forecast. It reduces overfitting and produces accurate projections.

Gradient Boosting:

Gradient Boosting is a boosting algorithm that iteratively improves weak learners by minimizing the loss function. It combines multiple weak learners into a strong learner to make accurate predictions.

XG Boost:

XG Boost is an optimized implementation of the Gradient Boosting algorithm. It incorporates regularization techniques and parallel processing to enhance performance and achieve high prediction accuracy.

3.5.3 DL Models:

Artificial neural networks with several layers, or "deep learning models," have the ability to detect and pick up complex patterns on their own. In this project, the DL model displayed below was applied:

MLP:

MLP is a feedforward neural network that features several hidden layers. As a result of its ability to learn non-linear correlations between input and output data, it is suited for difficult categorization problems.

3.6 Evaluation and Visualization:

3.6.1 Hyper Parameter Tuning:

The process of choosing a machine learning algorithm's ideal hyperparameter configuration is known as tuning. Hyperparameters are variables that are chosen before the model is trained and cannot be discovered by data learning.

To improve the performance of several machine learning models, such as classification and ensemble models as well as deep learning models, hyperparameter tweaking was done. Finding the ideal set of hyperparameters to maximise the performance and generalizability of a model is the goal of hyperparameter tweaking.

To improve the effectiveness of the classification model, hyperparameter tuning was done using the GridSearchCV approach. GridSearchCV painstakingly traverses a preset grid of hyperparameters, testing the model's performance via cross-validation.

- learning rate
- number of estimators
- maximum depth
- subsample
- colsample bytree

were among the hyperparameters taken into account for tuning. The ideal combination was found by adjusting these hyperparameters and assessing the model's precision. This procedure makes the fact that the model is optimised for the optimum performance on the provided dataset.

A collection of important hyperparameters was taken into account for each model, varying according to the particular method. These hyperparameters regulate a variety of elements involved in learning, including the model's complexity, regularisation, and optimisation technique. The learning rate, regularisation intensity, number of hidden layers, number of neurons per layer, and activation functions are a few examples of hyperparameters that are often modified.

For each hyperparameter, a range of values was investigated throughout the tuning process. GridSearchCV or random search methods were used to undertake a systematic search over the hyperparameter space and assess the performance of the models using suitable evaluation measures like accuracy, precision, recall, or F1-score. Based on these measurements, the ideal set of hyperparameters was chosen.

By contrasting the efficiency of the models prior to and following tuning, the effect of hyperparameter tweaking was evaluated. The chosen hyperparameters allowed the models to better capture complicated patterns and generalise effectively to new data, which resulted in the tweaked models performing better.

3.6.2 Evaluation Metrics:

Utilising assessment parameters such as accuracy, precision, recall, and F1 score, the effectiveness of each model was evaluated. The efficacy of each model was assessed using assessment criteria such AUC (Area Under the Curve) values derived from the ROC (Receiver Operating Characteristic) curve. Confusion matrices were made in order to assess how well the models worked for each sort of crime.

Accuracy:

The total accuracy of the model's predictions is measured by accuracy. It determines the proportion of accurately predicted occurrences to all of the dataset's instances.

Precision:

Out of all cases that are projected to be positive, precision is the percentage of accurately predicted positive instances. It shows how well the model can avoid producing false positives.

Recall:

The proportion of accurately anticipated positive cases out of all actual positive instances is measured by recall, also known as sensitivity or true positive rate. It shows how well the model can identify all instances of success.

F1 Score:

The harmonic mean of recall and precision is the F1 score. By taking into account both precision and recall, it offers a fair assessment of a model's performance.

Area Under the Curve (AUC):

For binary classification issues, AUC is a performance statistic. It reflects the region under the ROC curve, which shows how the ratio of true positives to false positives must be balanced. A model that performs better has a higher AUC value.

3.6.3 Visualization Dashboard:

Using Looker Studio, a dashboard was created that provides interactive visualisation of arrest vs. crime type and domestic vs. crime type. The dashboard has filters based on season and hourcategorized.

The visualization dashboard, created using Looker Studio, provides an interactive interface to explore crime patterns. Users can analyze the relationship between crime type Vs domestic violence and arrest. The dashboard allows filtering based on season and hourcategorized, enabling users to investigate crime trends during specific time periods. This visual representation enhances understanding and facilitates data-driven decision-making in crime prevention efforts.

Steps employed in this project is in Figure 3.4

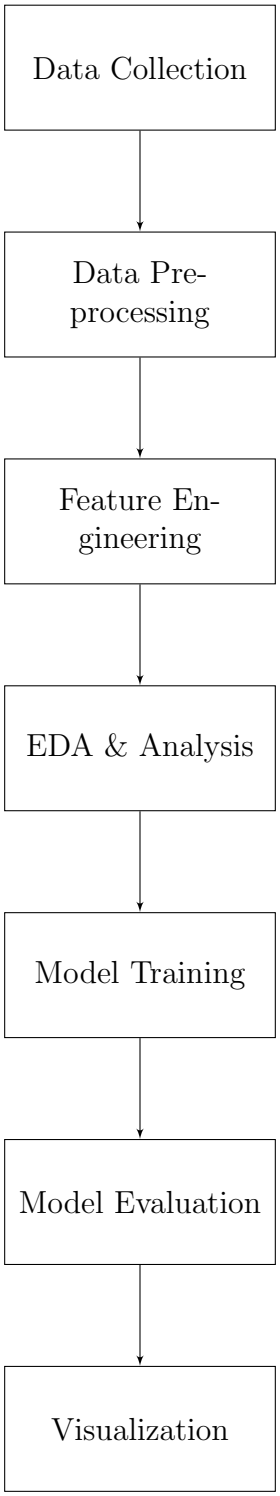

Figure 3.4: Project Workflow

Chapter 4

Results and Evaluation

4.1 EDA Findings:

The analysis of each crime category showed that drugs, theft, battery, and criminal property damage were the most common types. According to the distribution of crimes by hour, there was more activity in the afternoon and evening. There was no discernible periodicity in the examination of crime incidents by season. Different crime types had different relationships with arrests and domestic violence occurrences, with certain categories having greater arrest rates as well as domestic violence participation.

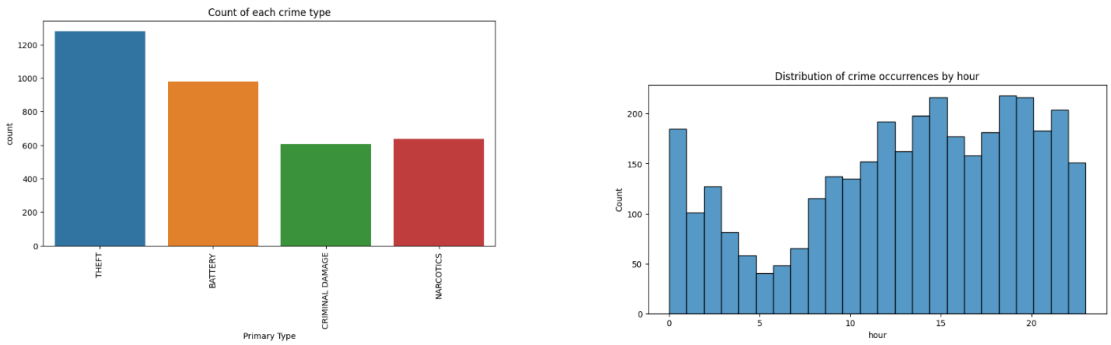

Figure 4.1: Count of each crime type (left) and Distribution of crime occurrences by hour (right)

4.2 Machine Learning Results:

The accuracy of each machine learning model was used as the key criterion to assess its performance. Among the examined models, XG Boost had the accuracy

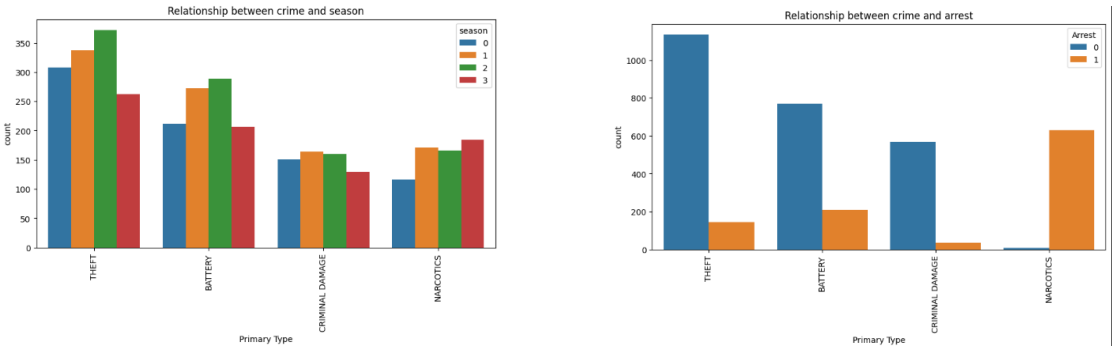

Figure 4.2: Distribution of crime occurrences by season (left) and Relationship between crime and arrest (right)

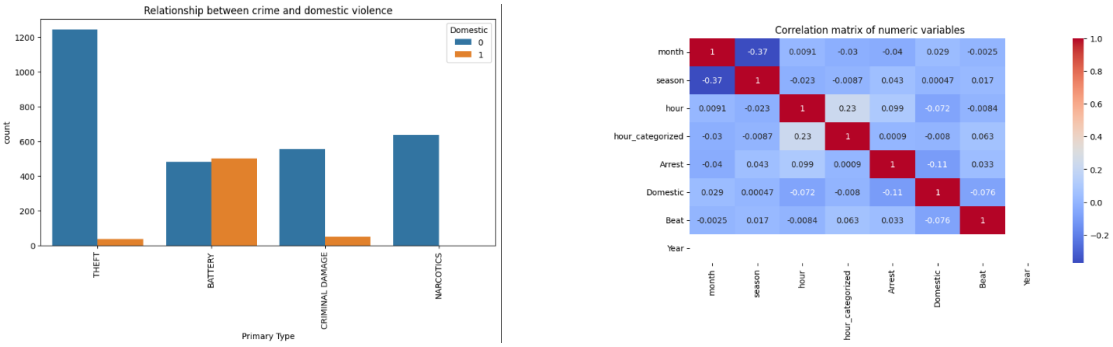

Figure 4.3: Relationship between crime and domestic violence (left) and Correlation matrix of numeric variables (right)

that was the greatest at 81. In addition to producing competitive accuracies, Random Forest, Logistic Regression, as well as SVM also show how well ensemble and conventional models predict crime.

Important insights into the performance of various models are revealed through the examination of the AUC values from the ROC curve and the accuracies for each class.

- The AUC values for Classes 0, 1, 2, and 3 for the SVM model are 0.72, 0.57, 0.96, and 0.73, respectively.
- Similar outcomes were obtained using the Random Forest and Logistic Regression models, with AUC values for all classes ranging from 0.76 to 0.77.
- Performance for the XG Boost model was somewhat better, with AUC values between 0.78 and 0.79.

In terms of accuracy,

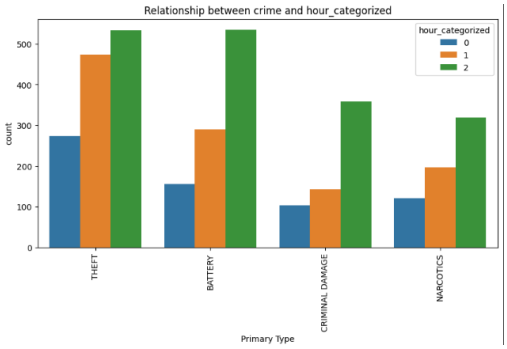

Figure 4.4: Relationship between crime and hour_categorized

- The Random Forest model had an accuracy of 0.747
- XG Boost model had the best accuracy of 0.81.
- Additionally, Logistic Regression performed admirably, with an accuracy of 0.694.
- The accuracy ranged from 0.587 to 0.63 for the KNN, SVM, Naive Bayes, Gradient Boost, SVM with Kernel Trick, and Multi-Layer Perceptron models, whereas these models showed lesser accuracy.

Table 4.1: Performance Metrics of XG Boost

Class	Precision	Recall	F1-score	Support
BATTERY	0.825	0.560	0.670	210
CRIMINAL DAMAGE	0.320	0.135	0.190	110
NARCOTICS	0.780	0.910	0.840	120
THEFT	0.590	0.830	0.690	250

Table 4.2: Performance Metrics of Random Forest

Class	Precision	Recall	F1-score	Support
BATTERY	0.779	0.517	0.622	205
CRIMINAL DAMAGE	0.265	0.108	0.154	120
NARCOTICS	0.741	0.922	0.822	115
THEFT	0.556	0.796	0.655	260

These findings lead to the conclusion that the XG Boost model performed the best overall, outperforming the other models in terms of accuracy and AUC. The

performance of the Random Forest and Logistic Regression models was similarly competitive.

These results demonstrate the potential of ensemble-based algorithms, such as XG Boost, for analysis and prediction of criminal activity. To increase the effectiveness and generalizability of the models, however, further research and testing may be necessary.

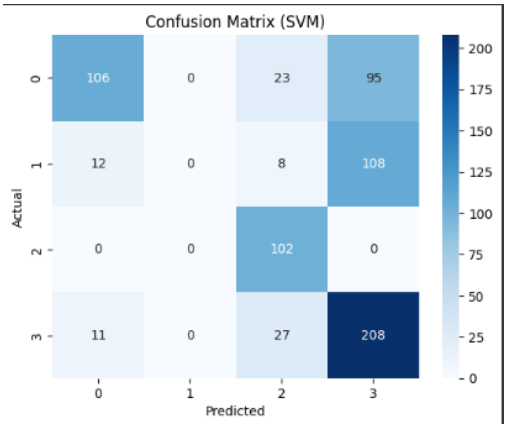

Figure 4.5: Confusion Matrix (SVM)

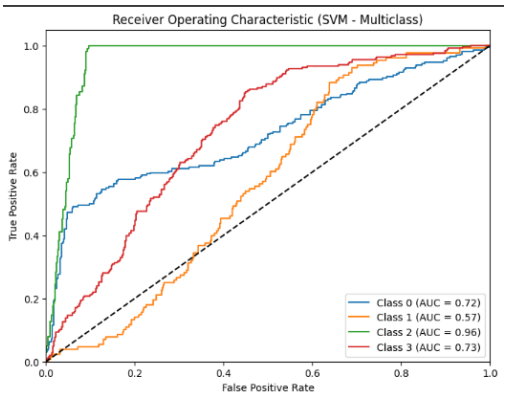

Figure 4.6: Receiver Operating Characteristic (SVM - Multiclass)

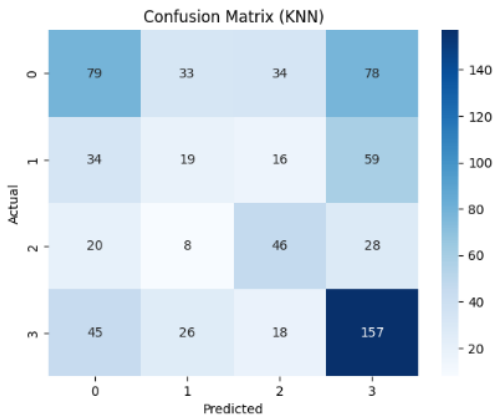

Figure 4.7: Confusion Matrix (KNN)

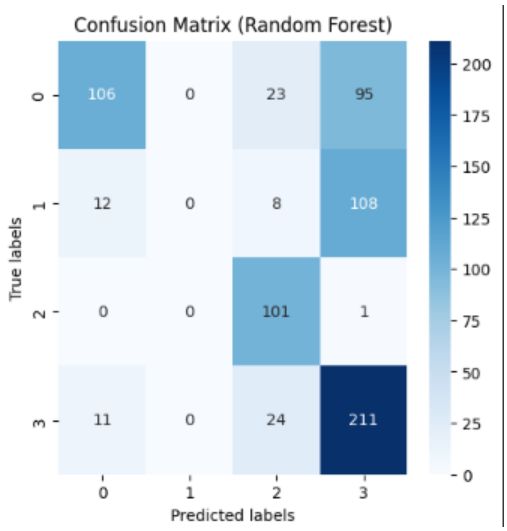

Figure 4.8: Confusion Matrix (Random Forest)

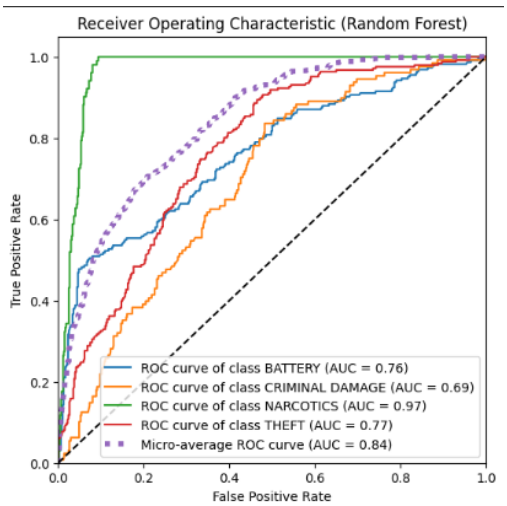

Figure 4.9: Receiver Operating Characteristic (Random Forest)

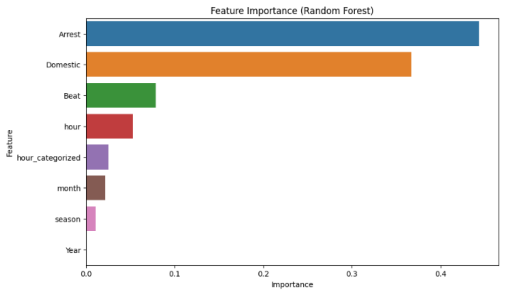

Figure 4.10: Feature Importance (Random Forest)

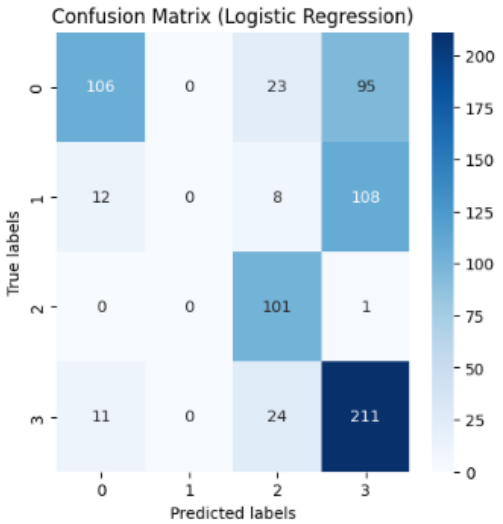

Figure 4.11: Confusion Matrix (Logistic Regression)

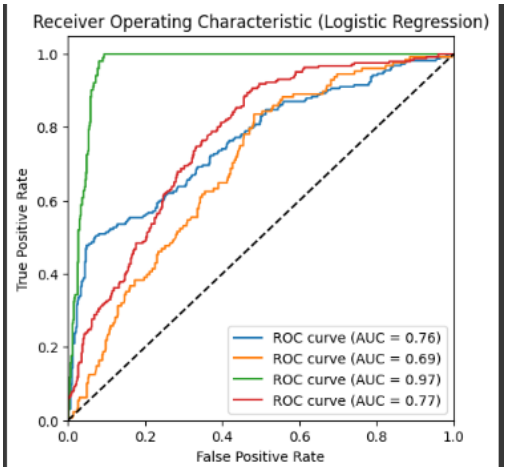

Figure 4.12: Receiver Operating Characteristic (Logistic Regression)

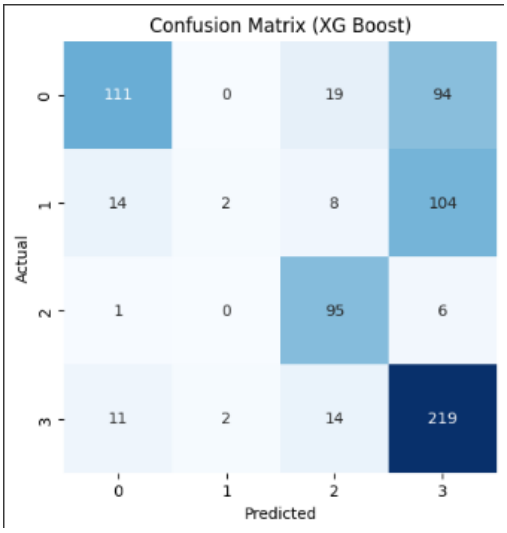

Figure 4.13: Confusion Matrix (XG Boost)

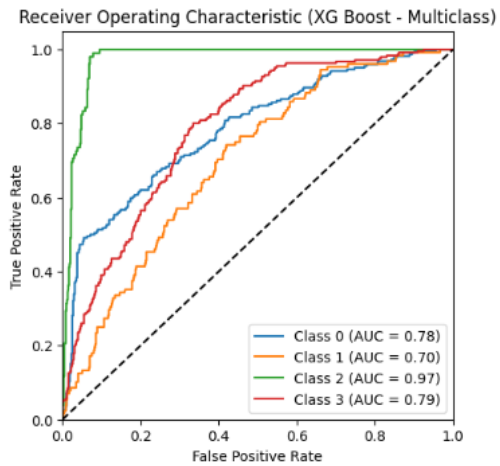

Figure 4.14: Receiver Operating Characteristic (XG Boost - Multiclass)

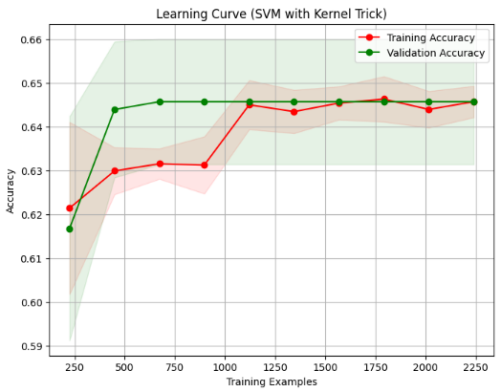

Figure 4.15: Learning Curve (SVM with Kernel Trick)

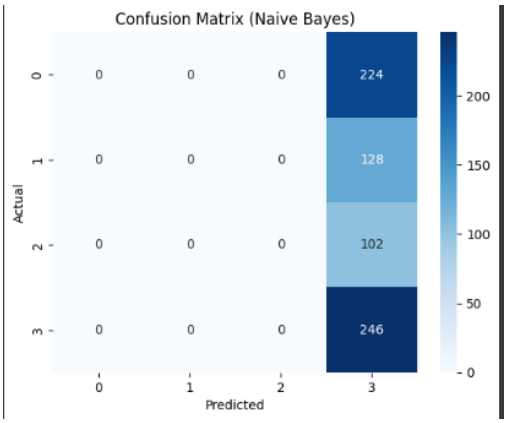

Figure 4.16: Confusion Matrix (Naive Bayes)

Table 4.3: Accuracies for each model

Model	Accuracy
XG Boost	0.810000
Random Forest	0.747143
Logistic Regression	0.694286
KNN	0.630000
SVM	0.624286
Naive Bayes	0.619071
Gradient Boost	0.615714
SVM with Kernel Trick	0.594286
Multi-Layer Perceptron	0.587143

Table 4.4: AUC values for each model

Model	Class0	Class1	Class2	Class3
SVM	0.72	0.57	0.96	0.73
Random Forest	0.76	0.69	0.97	0.77
Logistic Regression	0.76	0.69	0.97	0.77
XG Boost	0.78	0.70	0.97	0.79

4.3 Hot-Spot Mapping

A useful method for locating places with a lot of criminal activity is hot spot mapping, which is utilised in crime analysis. Hot spots may be found by projecting criminal occurrences on a map and examining spatial trends. These hot spots show where crime is concentrated. For law enforcement organisations to efficiently allocate resources and carry out targeted actions, this information is essential. In order to better understand the spatial distribution of crime and to design proactive crime prevention tactics, hot spot mapping provides a visual depiction of crime trends.

4.4 Visualization and Dashboard:

The project includes a visualisation dashboard created with Looker Studio, providing an easy-to-use interface for investigating crime trends and their relationships. Users of the dashboard may see how different crime kinds, arrests, and instances of domestic violence are related to the season and hourcategorized.

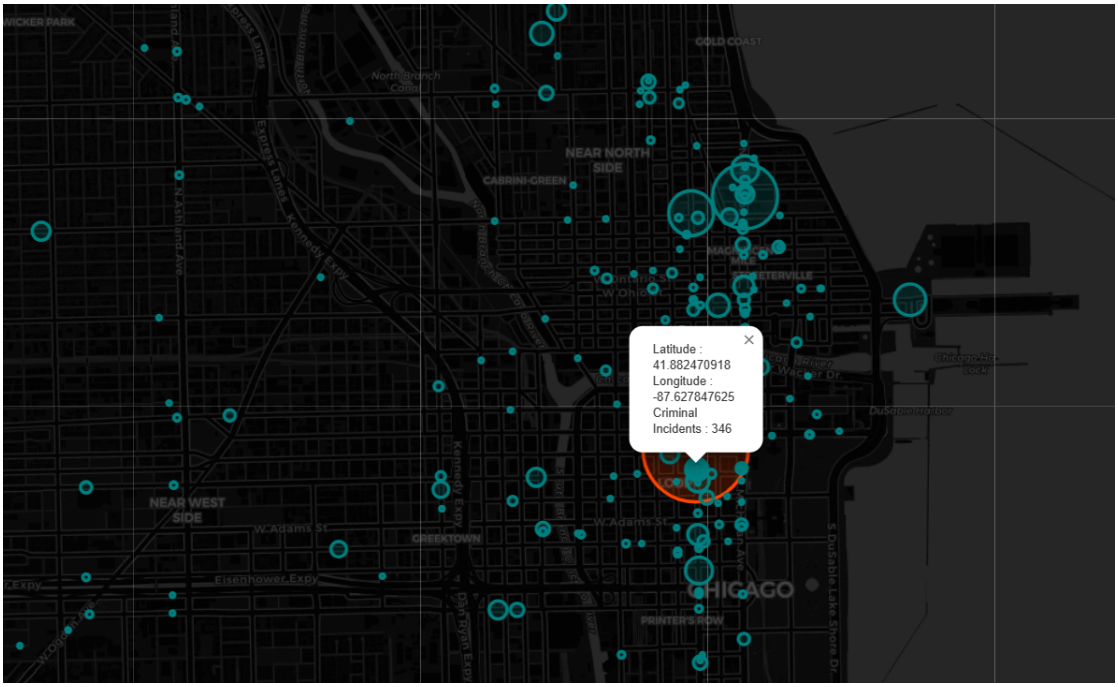

Figure 4.17: Hot Spot Mapping

Users of the dashboard may discover temporal patterns, investigate the effects of variables like arrests as well as domestic violence events, and receive useful insights into the spread of crime occurrences across various crime kinds. The visualization capabilities provide a thorough comprehension of criminal trends, assisting with resource allocation and reasoned decision-making.

The dashboard provides police agencies and other stakeholders with an effective and intuitive platform for exploring, analysing, and interpreting the intricate linkages contained within the crime dataset by utilising the power of data visualisation. Users are able to dig down into certain crime categories, time periods, and geographic locations using the dashboard's interactive features, which allows for a deeper knowledge of crime trends and the creation of tailored crime prevention measures.

In general, the project's dashboard and visualisation components improve the usability and accessibility of the crime data, empowering stakeholders to make deft choices based on data-driven insights. It is a useful tool for investigating crime trends, locating crime hotspots, and comprehending the nature of crimes occurring in the specified region and time frame.

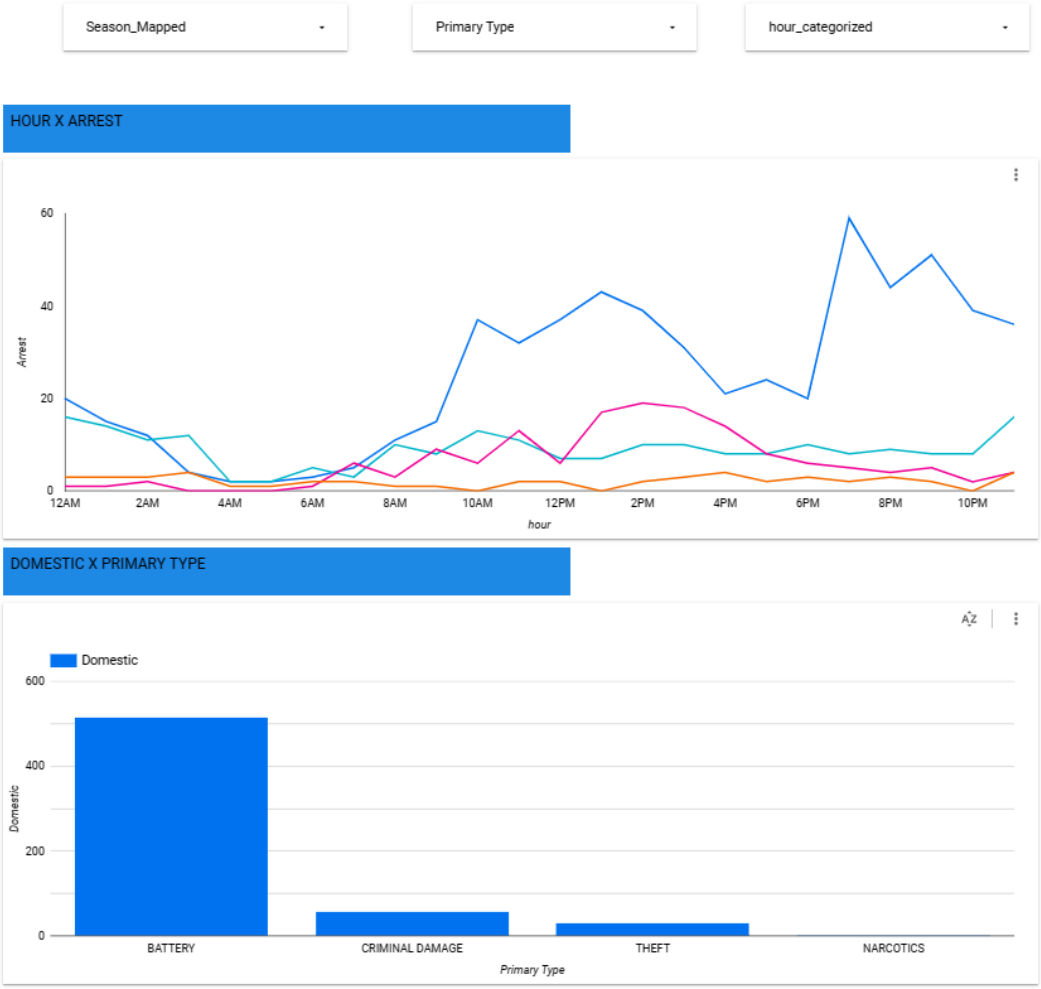

Figure 4.18: Looker Studio Dashboard

Chapter 5

FUTURE WORKS

5.1 Incorporating Additional Data:

The main goal of future study should be to include more data sources, which include social media data or sensor data, socioeconomic variables, meteorological data, and demographic data. By including such data, it may be possible to gain a deeper knowledge of crime trends and improve the precision of the models used for prediction.

5.2 Advanced Techniques and Algorithms:

The models' performance may be further enhanced by investigating cutting-edge feature engineering strategies, dimensionality reduction techniques, and anomaly detection algorithms, which would also make it possible to identify rare or atypical crime patterns.

In addition, further exploration of deep learning techniques could be beneficial in improving the accuracy and robustness of the crime prediction models. Deep learning models such as recurrent neural networks (RNNs) or convolutional neural networks (CNNs) can capture complex patterns and dependencies within the data.

5.3 Real-time Implementation:

The integration of the created models into systems that forecast crime in real-time should be a future goal. The difficulties and necessities of real-world implementation, including scalability, computational effectiveness, and data privacy, should be taken into account.

Chapter 6

Conclusion

6.1 Summary of Results:

In conclusion, our project successfully employed machine learning techniques, specifically ensemble learning, to develop accurate crime prediction models. The models demonstrated strong performance in classifying different types of crimes based on historical data. Exploratory data analysis provided valuable insights into temporal trends, crime linkages, and patterns.

6.2 Contributions:

This project contributes to the field of crime analysis and prediction by providing law enforcement authorities with effective tools for informed decision-making. The utilization of ensemble learning techniques improved the accuracy and reliability of crime prediction models. Additionally, the visualization dashboard offers a user-friendly platform for examining crime trends and related factors.

6.3 Practical Implications:

In terms of resource allocation, crime prevention tactics, and decision-making procedures, the project's findings have immediate applications for law enforcement organisations. Utilising trustworthy crime prediction models, authorities may focus their efforts on regions with greater rates of criminal activity and allocate resources more efficiently. Improved public safety and more effective crime prevention may result from this.

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